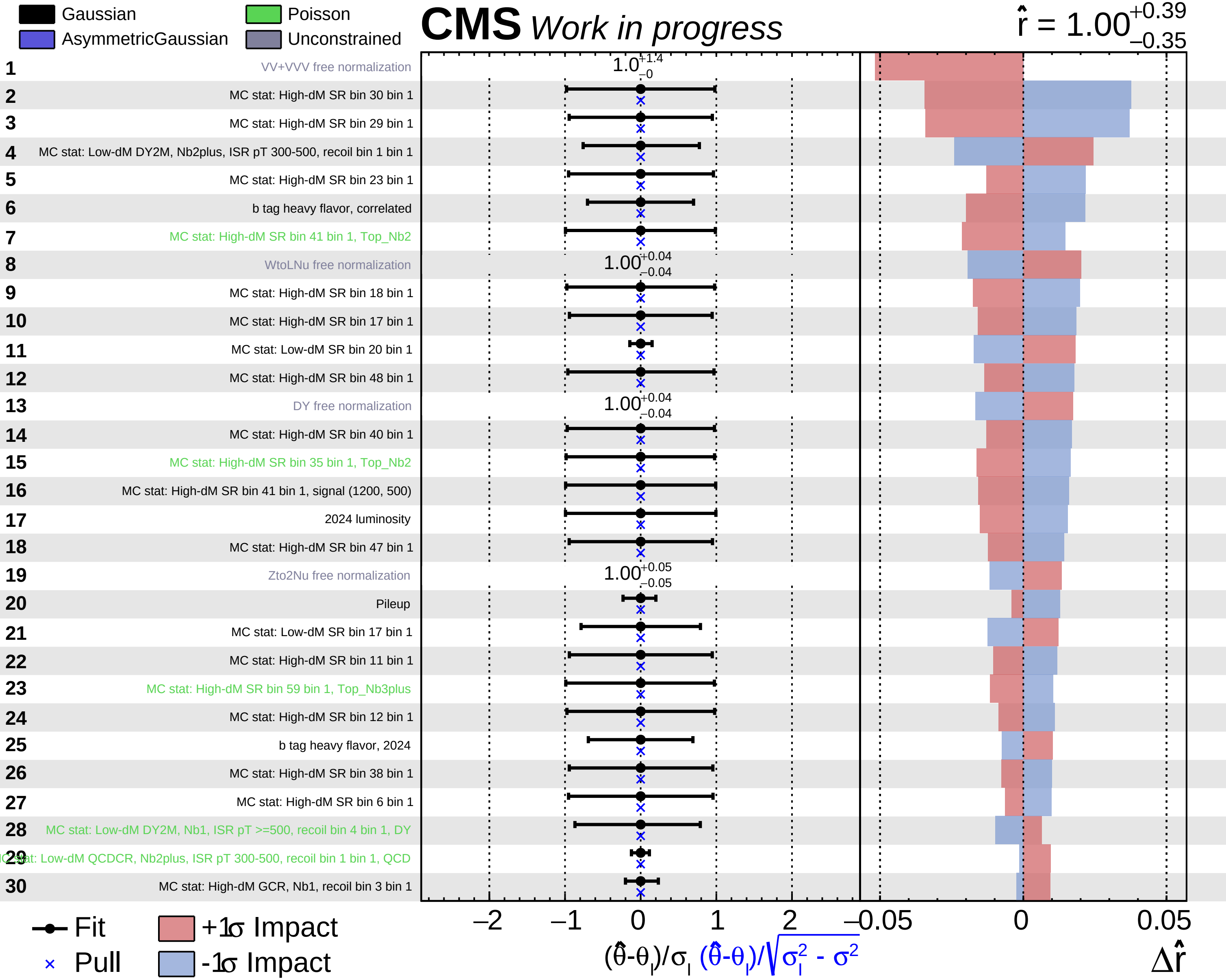
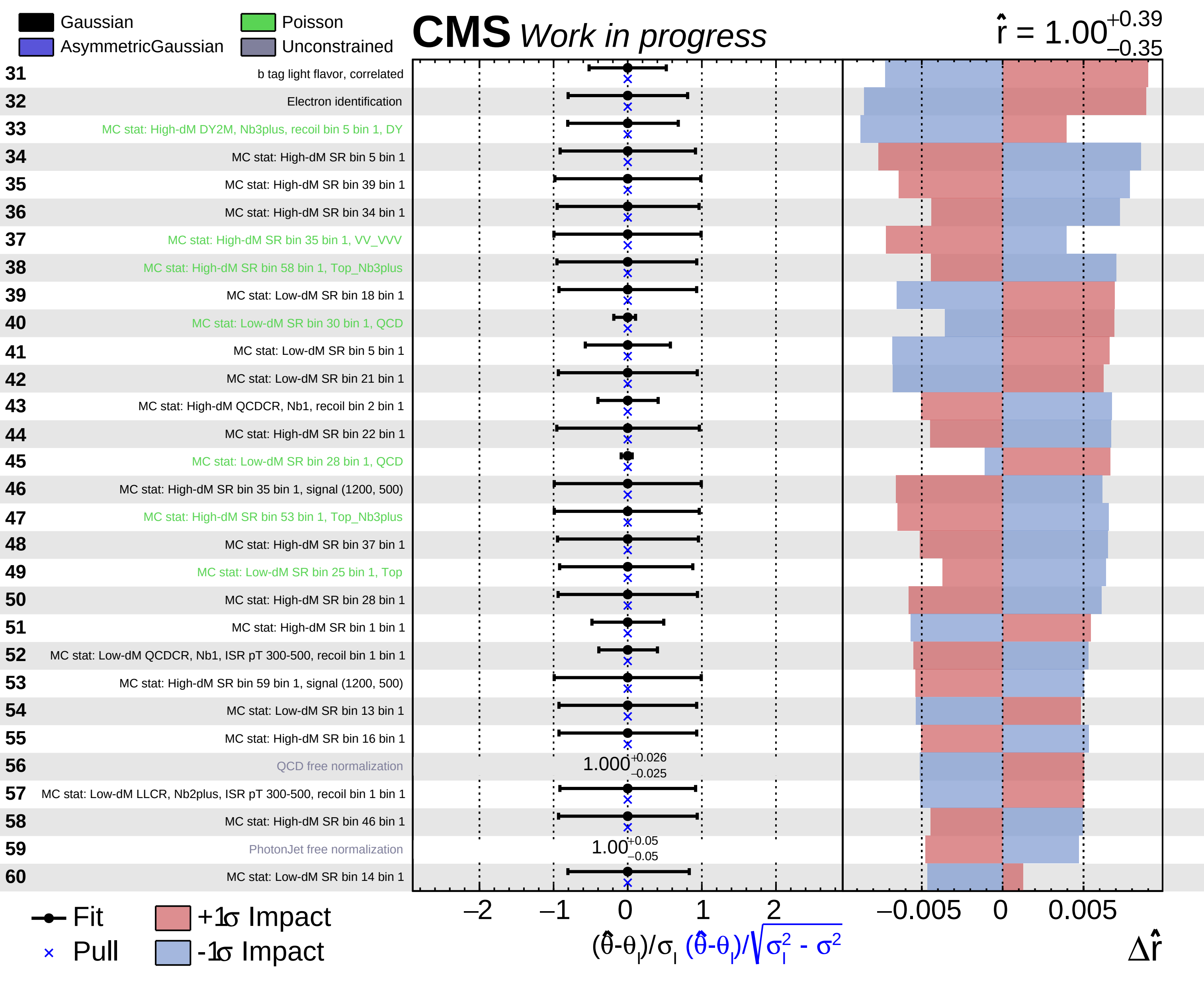
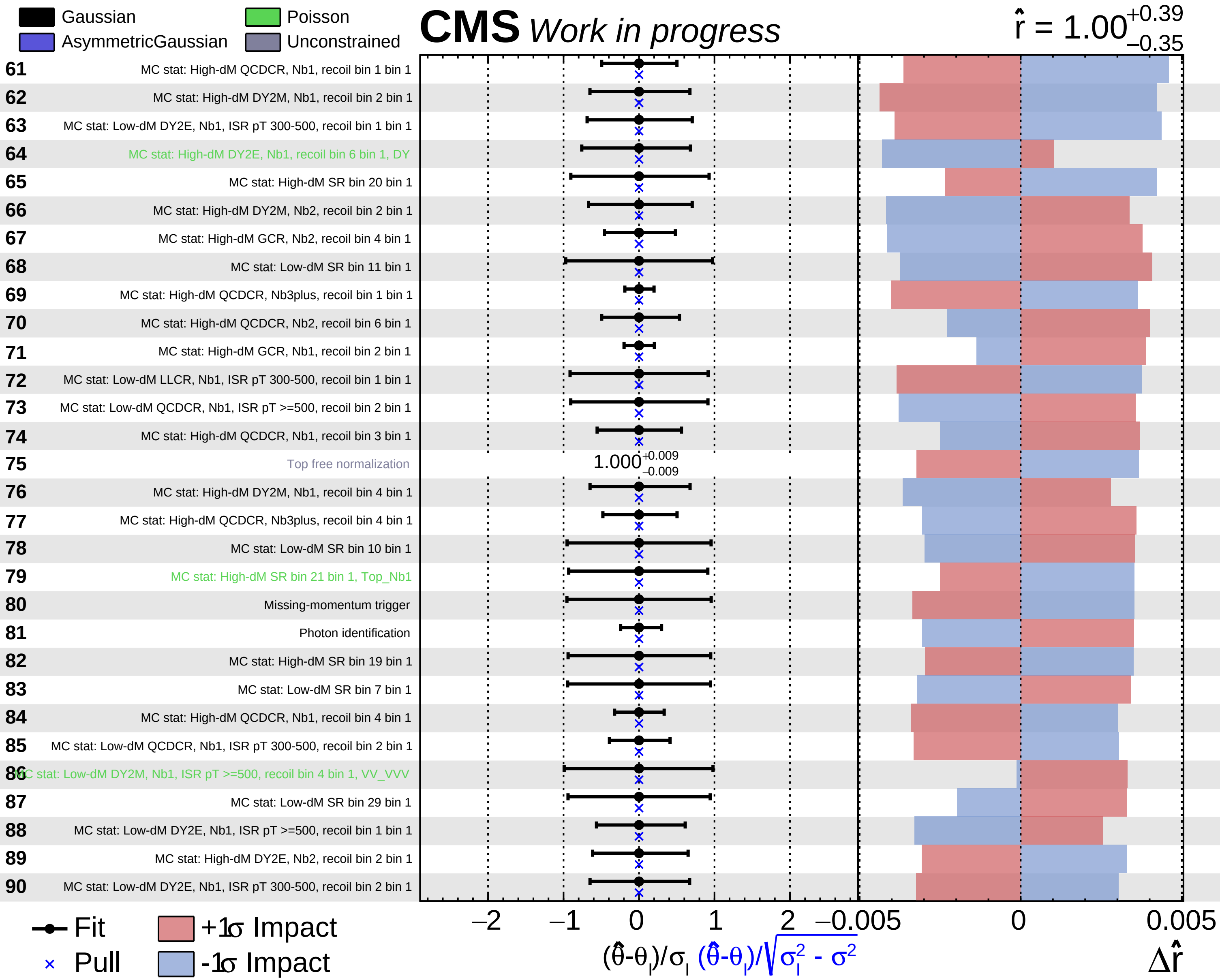
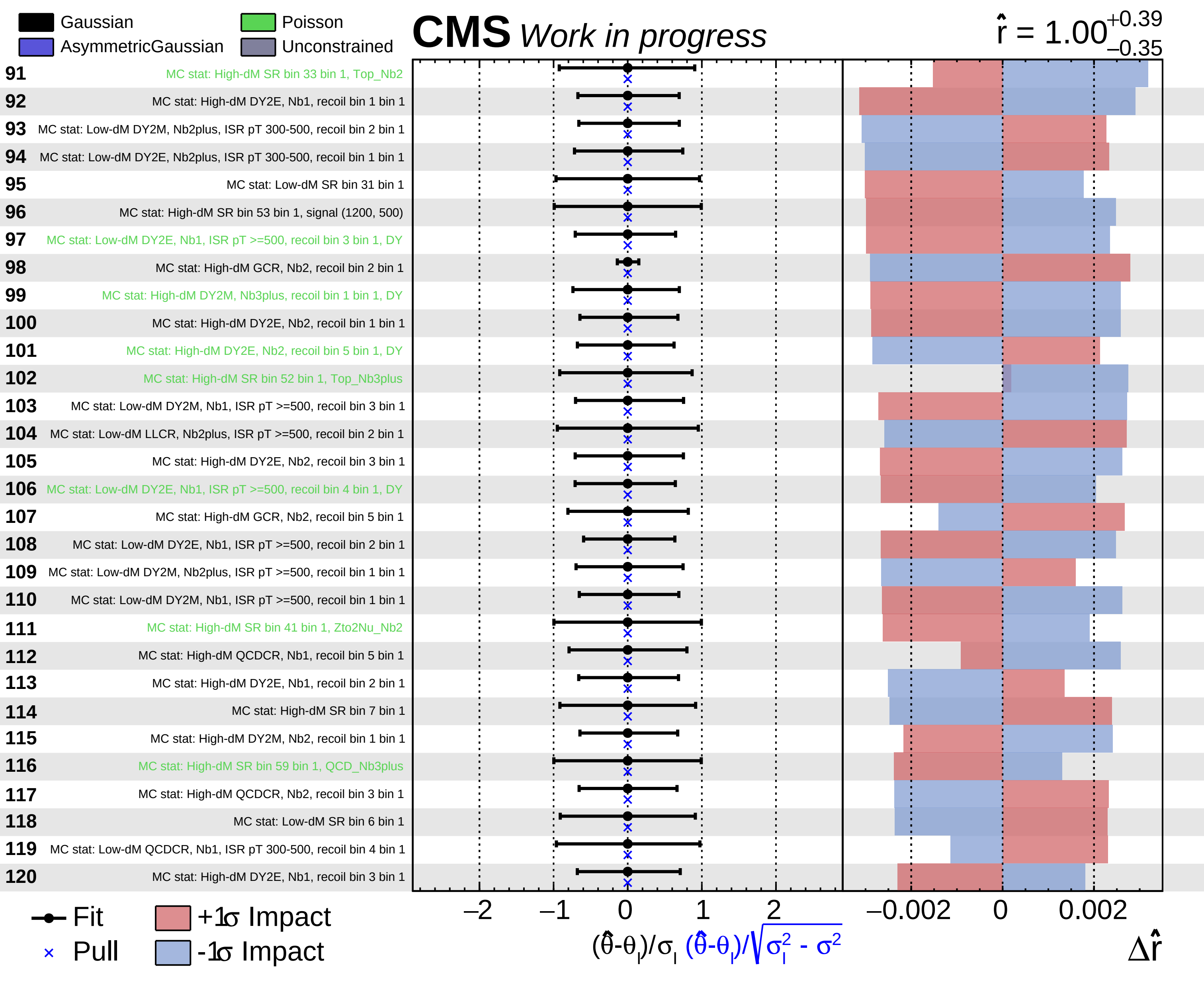

Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.39 −0.35
1 VV+VVV free normalization
1.0 +1.4 −0
2 MC stat: High-dM SR bin 30 bin 1
3 MC stat: High-dM SR bin 29 bin 1
4 MC stat: Low-dM DY2M, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1
5 MC stat: High-dM SR bin 23 bin 1
6 b tag heavy flavor, correlated
7 MC stat: High-dM SR bin 41 bin 1, Top_Nb2
8 WtoLNu free normalization
1.00 +0.04 −0.04
9 MC stat: High-dM SR bin 18 bin 1
10 MC stat: High-dM SR bin 17 bin 1
11 MC stat: Low-dM SR bin 20 bin 1
12 MC stat: High-dM SR bin 48 bin 1
13 DY free normalization
1.00 +0.04 −0.04
14 MC stat: High-dM SR bin 40 bin 1
15 MC stat: High-dM SR bin 35 bin 1, Top_Nb2
16 MC stat: High-dM SR bin 41 bin 1, signal (1200, 500)
17 2024 luminosity
18 MC stat: High-dM SR bin 47 bin 1
19 Zto2Nu free normalization
1.00 +0.05 −0.05
20 Pileup
21 MC stat: Low-dM SR bin 17 bin 1
22 MC stat: High-dM SR bin 11 bin 1
23 MC stat: High-dM SR bin 59 bin 1, Top_Nb3plus
24 MC stat: High-dM SR bin 12 bin 1
25 b tag heavy flavor, 2024
26 MC stat: High-dM SR bin 38 bin 1
27 MC stat: High-dM SR bin 6 bin 1
28 MC stat: Low-dM DY2M, Nb1, ISR pT >=500, recoil bin 4 bin 1, DY
29 stat: Low-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, QCD
30 MC stat: High-dM GCR, Nb1, recoil bin 3 bin 1
−2 −1 0 1 2
−0.05 0 0.05
Fit
Pull
+1σ Impact
−1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

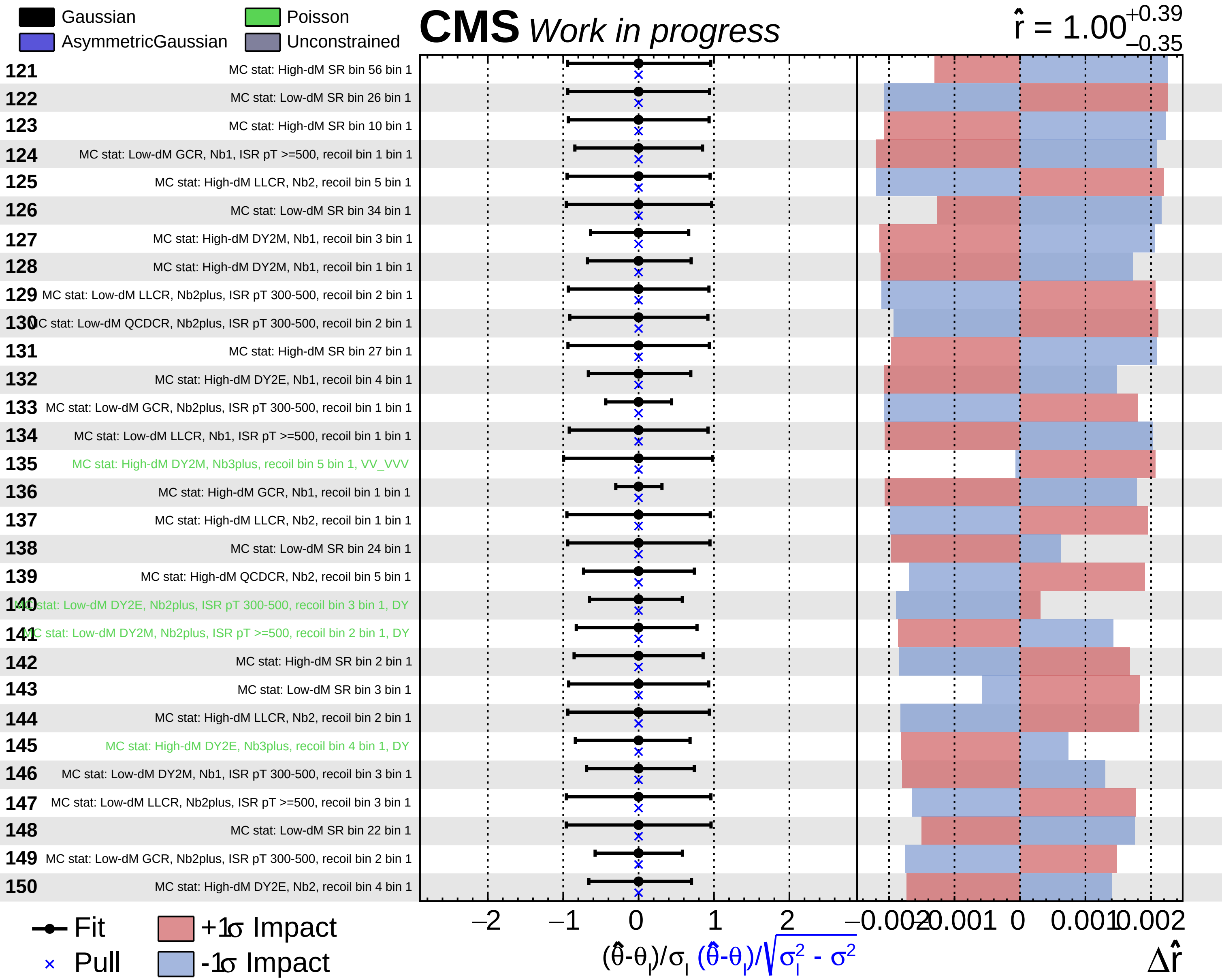


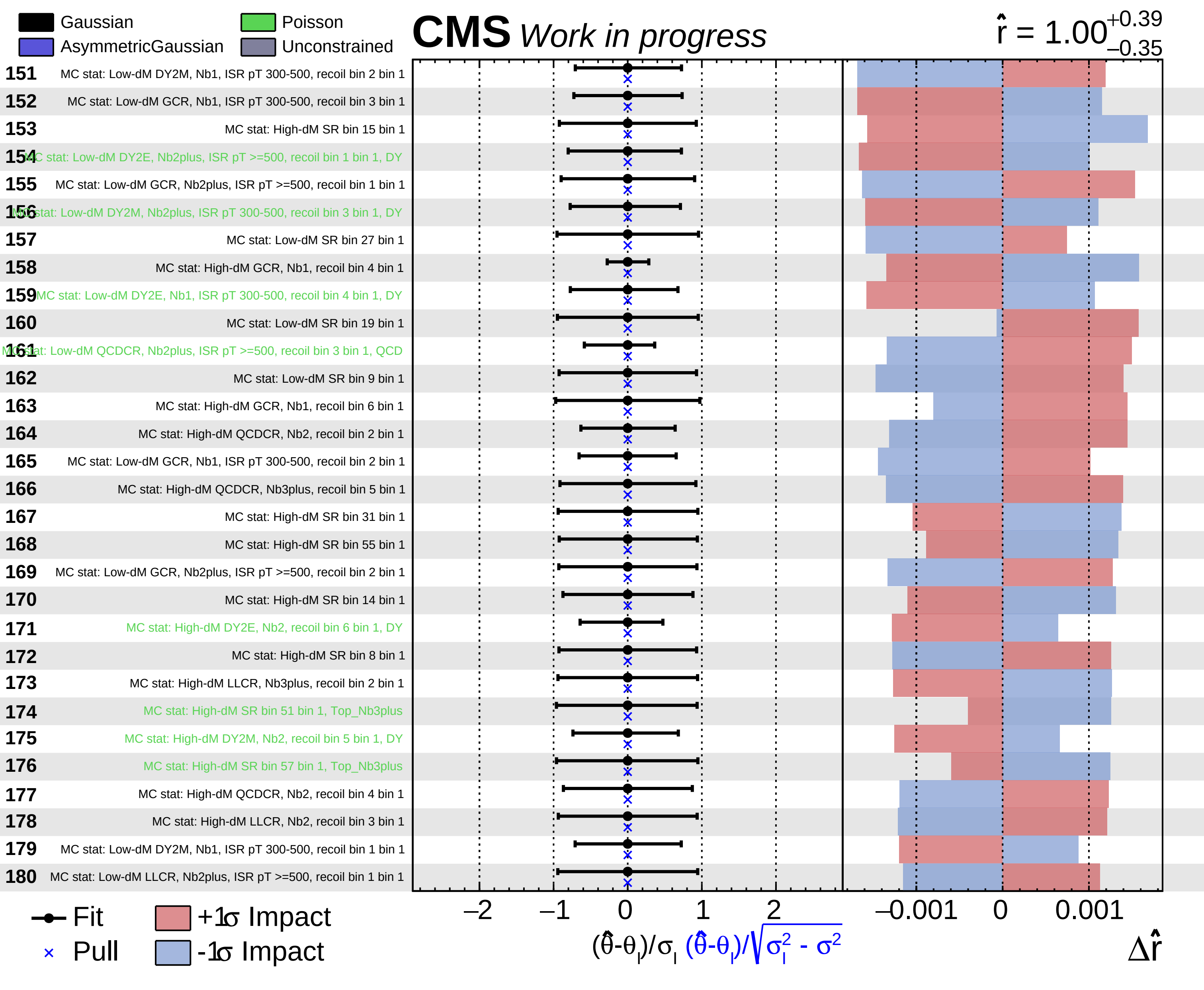
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
31 b tag light flavor, correlated
32 Electron identification
33 MC stat: High-dM DY2M, Nb3plus, recoil bin 5 bin 1, DY
34 MC stat: High-dM SR bin 5 bin 1
35 MC stat: High-dM SR bin 39 bin 1
36 MC stat: High-dM SR bin 34 bin 1
37 MC stat: High-dM SR bin 35 bin 1, VV_VVV
38 MC stat: High-dM SR bin 58 bin 1, Top_Nb3plus
39 MC stat: Low-dM SR bin 18 bin 1
40 MC stat: Low-dM SR bin 30 bin 1, QCD
41 MC stat: Low-dM SR bin 5 bin 1
42 MC stat: Low-dM SR bin 21 bin 1
43 MC stat: High-dM QCDCR, Nb1, recoil bin 2 bin 1
44 MC stat: High-dM SR bin 22 bin 1
45 MC stat: Low-dM SR bin 28 bin 1, QCD
46 MC stat: High-dM SR bin 35 bin 1, signal (1200, 500)
47 MC stat: High-dM SR bin 53 bin 1, Top_Nb3plus
48 MC stat: High-dM SR bin 37 bin 1
49 MC stat: Low-dM SR bin 25 bin 1, Top
50 MC stat: High-dM SR bin 28 bin 1
51 MC stat: High-dM SR bin 1 bin 1
52 MC stat: Low-dM QCDCR, Nb1, ISR pT 300-500, recoil bin 1 bin 1
53 MC stat: High-dM SR bin 59 bin 1, signal (1200, 500)
54 MC stat: Low-dM SR bin 13 bin 1
55 MC stat: High-dM SR bin 16 bin 1
56 QCD free normalization
$1.000^{+0.026}_{-0.025}$
57 MC stat: Low-dM LLCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1
58 MC stat: High-dM SR bin 46 bin 1
59 PhotonJet free normalization
$1.00^{+0.05}_{-0.05}$
60 MC stat: Low-dM SR bin 14 bin 1
−2
−1
0
1
2
−0.005
0
0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

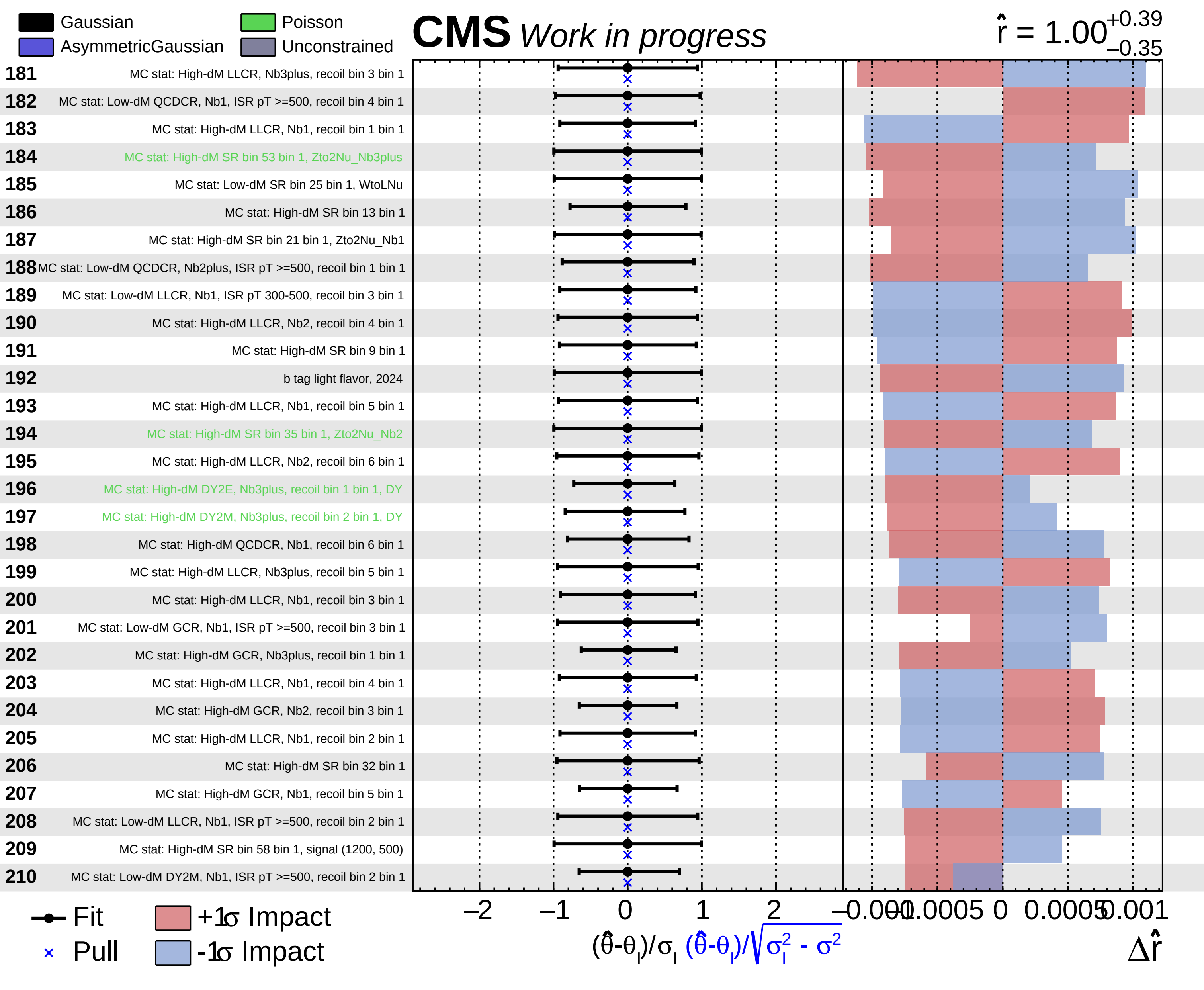


Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.39 −0.35
61 MC stat: High-dM QCDCR, Nb1, recoil bin 1 bin 1
62 MC stat: High-dM DY2M, Nb1, recoil bin 2 bin 1
63 MC stat: Low-dM DY2E, Nb1, ISR pT 300-500, recoil bin 1 bin 1
64 MC stat: High-dM DY2E, Nb1, recoil bin 6 bin 1, DY
65 MC stat: High-dM SR bin 20 bin 1
66 MC stat: High-dM DY2M, Nb2, recoil bin 2 bin 1
67 MC stat: High-dM GCR, Nb2, recoil bin 4 bin 1
68 MC stat: Low-dM SR bin 11 bin 1
69 MC stat: High-dM QCDCR, Nb3plus, recoil bin 1 bin 1
70 MC stat: High-dM QCDCR, Nb2, recoil bin 6 bin 1
71 MC stat: High-dM GCR, Nb1, recoil bin 2 bin 1
72 MC stat: Low-dM LLCR, Nb1, ISR pT 300-500, recoil bin 1 bin 1
73 MC stat: Low-dM QCDCR, Nb1, ISR pT >=500, recoil bin 2 bin 1
74 MC stat: High-dM QCDCR, Nb1, recoil bin 3 bin 1
75 Top free normalization
1.000 +0.009 −0.009
76 MC stat: High-dM DY2M, Nb1, recoil bin 4 bin 1
77 MC stat: High-dM QCDCR, Nb3plus, recoil bin 4 bin 1
78 MC stat: Low-dM SR bin 10 bin 1
79 MC stat: High-dM SR bin 21 bin 1, Top_Nb1
80 Missing-momentum trigger
81 Photon identification
82 MC stat: High-dM SR bin 19 bin 1
83 MC stat: Low-dM SR bin 7 bin 1
84 MC stat: High-dM QCDCR, Nb1, recoil bin 4 bin 1
85 MC stat: Low-dM QCDCR, Nb1, ISR pT 300-500, recoil bin 2 bin 1
86 MC stat: Low-dM DY2M, Nb1, ISR pT >=500, recoil bin 4 bin 1, VV_VVV
87 MC stat: Low-dM SR bin 29 bin 1
88 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 1 bin 1
89 MC stat: High-dM DY2E, Nb2, recoil bin 2 bin 1
90 MC stat: Low-dM DY2E, Nb1, ISR pT 300-500, recoil bin 2 bin 1
−2
−1
0
1
2
−0.005
0
0.005
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

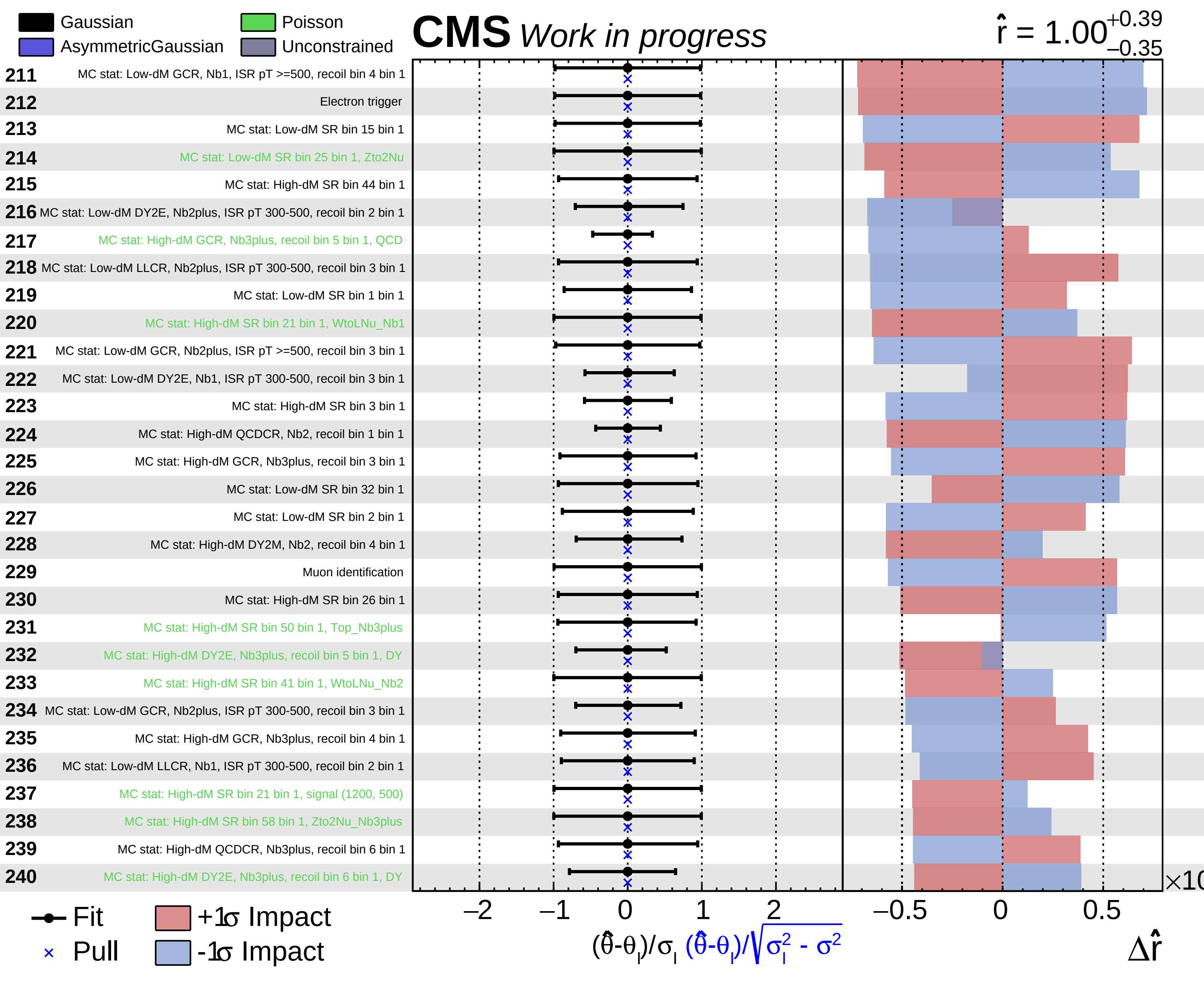

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
91 MC stat: High-dM SR bin 33 bin 1, Top_Nb2
92 MC stat: High-dM DY2E, Nb1, recoil bin 1 bin 1
93 MC stat: Low-dM DY2M, Nb2plus, ISR pT 300-500, recoil bin 2 bin 1
94 MC stat: Low-dM DY2E, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1
95 MC stat: Low-dM SR bin 31 bin 1
96 MC stat: High-dM SR bin 53 bin 1, signal (1200, 500)
97 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 3 bin 1, DY
98 MC stat: High-dM GCR, Nb2, recoil bin 2 bin 1
99 MC stat: High-dM DY2M, Nb3plus, recoil bin 1 bin 1, DY
100 MC stat: High-dM DY2E, Nb2, recoil bin 1 bin 1
101 MC stat: High-dM DY2E, Nb2, recoil bin 5 bin 1, DY
102 MC stat: High-dM SR bin 52 bin 1, Top_Nb3plus
103 MC stat: Low-dM DY2M, Nb1, ISR pT >=500, recoil bin 3 bin 1
104 MC stat: Low-dM LLCR, Nb2plus, ISR pT >=500, recoil bin 2 bin 1
105 MC stat: High-dM DY2E, Nb2, recoil bin 3 bin 1
106 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 4 bin 1, DY
107 MC stat: High-dM GCR, Nb2, recoil bin 5 bin 1
108 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 2 bin 1
109 MC stat: Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 1 bin 1
110 MC stat: Low-dM DY2M, Nb1, ISR pT >=500, recoil bin 1 bin 1
111 MC stat: High-dM SR bin 41 bin 1, Zto2Nu_Nb2
112 MC stat: High-dM QCDCR, Nb1, recoil bin 5 bin 1
113 MC stat: High-dM DY2E, Nb1, recoil bin 2 bin 1
114 MC stat: High-dM SR bin 7 bin 1
115 MC stat: High-dM DY2M, Nb2, recoil bin 1 bin 1
116 MC stat: High-dM SR bin 59 bin 1, QCD_Nb3plus
117 MC stat: High-dM QCDCR, Nb2, recoil bin 3 bin 1
118 MC stat: Low-dM SR bin 6 bin 1
119 MC stat: Low-dM QCDCR, Nb1, ISR pT 300-500, recoil bin 4 bin 1
120 MC stat: High-dM DY2E, Nb1, recoil bin 3 bin 1
−2
−1
0
1
2
−0.002
0
0.002
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.39 −0.35
121 MC stat: High-dM SR bin 56 bin 1
122 MC stat: Low-dM SR bin 26 bin 1
123 MC stat: High-dM SR bin 10 bin 1
124 MC stat: Low-dM GCR, Nb1, ISR pT >=500, recoil bin 1 bin 1
125 MC stat: High-dM LLCR, Nb2, recoil bin 5 bin 1
126 MC stat: Low-dM SR bin 34 bin 1
127 MC stat: High-dM DY2M, Nb1, recoil bin 3 bin 1
128 MC stat: High-dM DY2M, Nb1, recoil bin 1 bin 1
129 MC stat: Low-dM LLCR, Nb2plus, ISR pT 300-500, recoil bin 2 bin 1
130 MC stat: Low-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 2 bin 1
131 MC stat: High-dM SR bin 27 bin 1
132 MC stat: High-dM DY2E, Nb1, recoil bin 4 bin 1
133 MC stat: Low-dM GCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1
134 MC stat: Low-dM LLCR, Nb1, ISR pT >=500, recoil bin 1 bin 1
135 MC stat: High-dM DY2M, Nb3plus, recoil bin 5 bin 1, VV_VVV
136 MC stat: High-dM GCR, Nb1, recoil bin 1 bin 1
137 MC stat: High-dM LLCR, Nb2, recoil bin 1 bin 1
138 MC stat: Low-dM SR bin 24 bin 1
139 MC stat: High-dM QCDCR, Nb2, recoil bin 5 bin 1
140 MC stat: Low-dM DY2E, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, DY
141 MC stat: Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 2 bin 1, DY
142 MC stat: High-dM SR bin 2 bin 1
143 MC stat: Low-dM SR bin 3 bin 1
144 MC stat: High-dM LLCR, Nb2, recoil bin 2 bin 1
145 MC stat: High-dM DY2E, Nb3plus, recoil bin 4 bin 1, DY
146 MC stat: Low-dM DY2M, Nb1, ISR pT 300-500, recoil bin 3 bin 1
147 MC stat: Low-dM LLCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1
148 MC stat: Low-dM SR bin 22 bin 1
149 MC stat: Low-dM GCR, Nb2plus, ISR pT 300-500, recoil bin 2 bin 1
150 MC stat: High-dM DY2E, Nb2, recoil bin 4 bin 1
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Fit
Pull
+1σ Impact
−1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

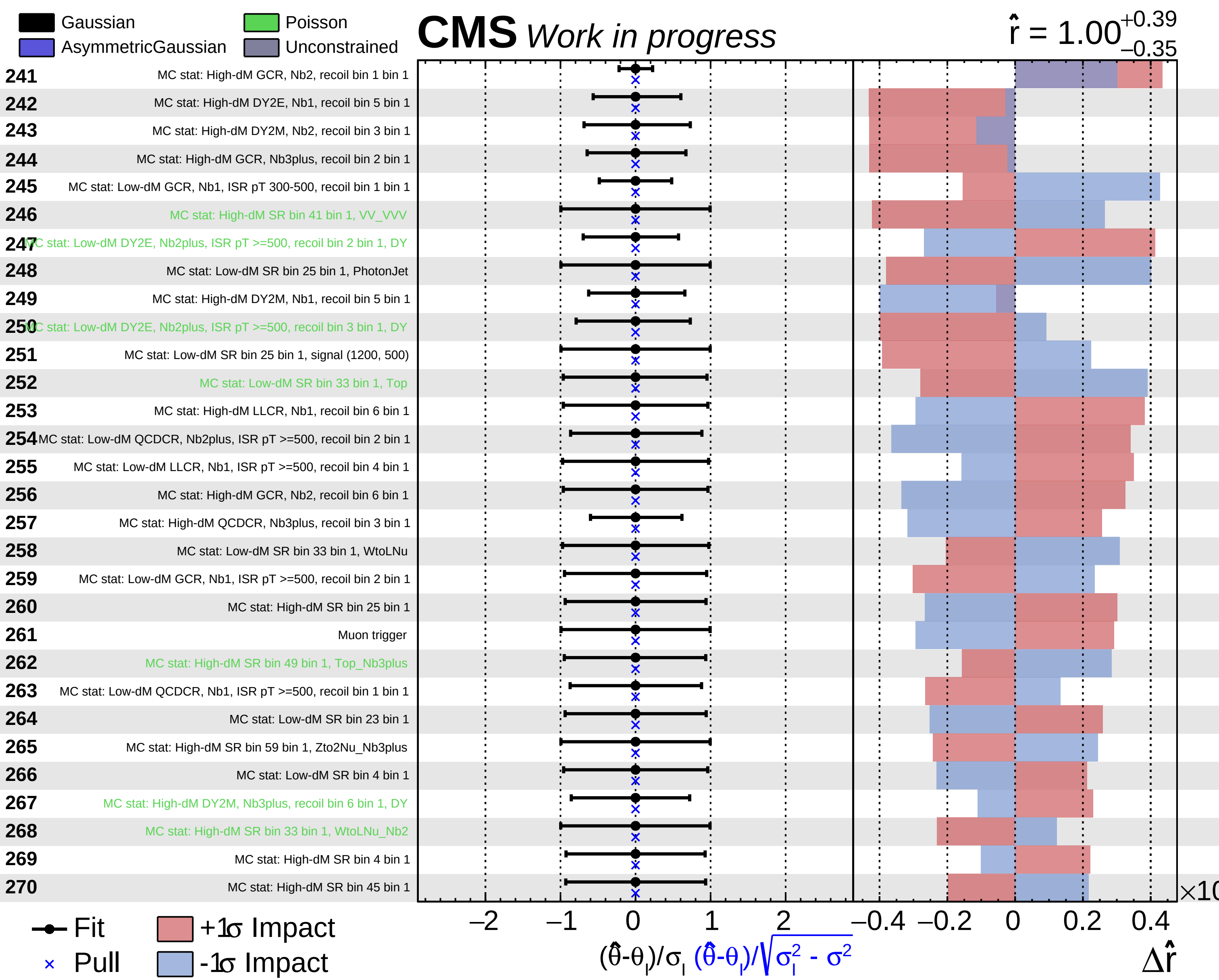

Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
151 MC stat: Low-dM DY2M, Nb1, ISR pT 300-500, recoil bin 2 bin 1
152 MC stat: Low-dM GCR, Nb1, ISR pT 300-500, recoil bin 3 bin 1
153 MC stat: High-dM SR bin 15 bin 1
154 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 1 bin 1, DY
155 MC stat: Low-dM GCR, Nb2plus, ISR pT >=500, recoil bin 1 bin 1
156 MC stat: Low-dM DY2M, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, DY
157 MC stat: Low-dM SR bin 27 bin 1
158 MC stat: High-dM GCR, Nb1, recoil bin 4 bin 1
159 MC stat: Low-dM DY2E, Nb1, ISR pT 300-500, recoil bin 4 bin 1, DY
160 MC stat: Low-dM SR bin 19 bin 1
161 MC stat: Low-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, QCD
162 MC stat: Low-dM SR bin 9 bin 1
163 MC stat: High-dM GCR, Nb1, recoil bin 6 bin 1
164 MC stat: High-dM QCDCR, Nb2, recoil bin 2 bin 1
165 MC stat: Low-dM GCR, Nb1, ISR pT 300-500, recoil bin 2 bin 1
166 MC stat: High-dM QCDCR, Nb3plus, recoil bin 5 bin 1
167 MC stat: High-dM SR bin 31 bin 1
168 MC stat: High-dM SR bin 55 bin 1
169 MC stat: Low-dM GCR, Nb2plus, ISR pT >=500, recoil bin 2 bin 1
170 MC stat: High-dM SR bin 14 bin 1
171 MC stat: High-dM DY2E, Nb2, recoil bin 6 bin 1, DY
172 MC stat: High-dM SR bin 8 bin 1
173 MC stat: High-dM LLCR, Nb3plus, recoil bin 2 bin 1
174 MC stat: High-dM SR bin 51 bin 1, Top_Nb3plus
175 MC stat: High-dM DY2M, Nb2, recoil bin 5 bin 1, DY
176 MC stat: High-dM SR bin 57 bin 1, Top_Nb3plus
177 MC stat: High-dM QCDCR, Nb2, recoil bin 4 bin 1
178 MC stat: High-dM LLCR, Nb2, recoil bin 3 bin 1
179 MC stat: Low-dM DY2M, Nb1, ISR pT 300-500, recoil bin 1 bin 1
180 MC stat: Low-dM LLCR, Nb2plus, ISR pT >=500, recoil bin 1 bin 1
−2
−1
0
1
2
−0.001
0
0.001
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



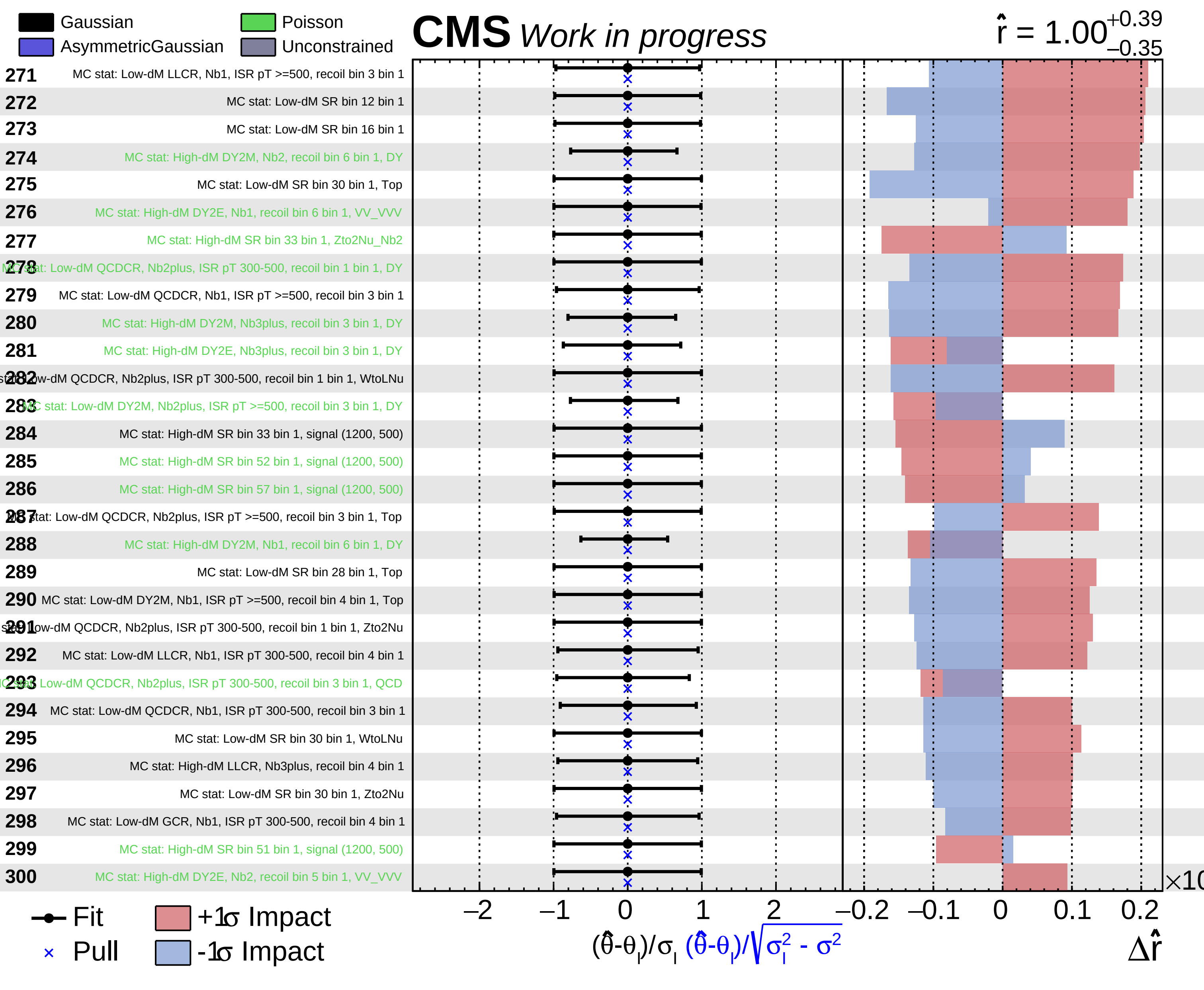
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
181 MC stat: High-dM LLCR, Nb3plus, recoil bin 3 bin 1
182 MC stat: Low-dM QCDCR, Nb1, ISR pT >=500, recoil bin 4 bin 1
183 MC stat: High-dM LLCR, Nb1, recoil bin 1 bin 1
184 MC stat: High-dM SR bin 53 bin 1, Zto2Nu_Nb3plus
185 MC stat: Low-dM SR bin 25 bin 1, WtoLNu
186 MC stat: High-dM SR bin 13 bin 1
187 MC stat: High-dM SR bin 21 bin 1, Zto2Nu_Nb1
188 MC stat: Low-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 1 bin 1
189 MC stat: Low-dM LLCR, Nb1, ISR pT 300-500, recoil bin 3 bin 1
190 MC stat: High-dM LLCR, Nb2, recoil bin 4 bin 1
191 MC stat: High-dM SR bin 9 bin 1
192 b tag light flavor, 2024
193 MC stat: High-dM LLCR, Nb1, recoil bin 5 bin 1
194 MC stat: High-dM SR bin 35 bin 1, Zto2Nu_Nb2
195 MC stat: High-dM LLCR, Nb2, recoil bin 6 bin 1
196 MC stat: High-dM DY2E, Nb3plus, recoil bin 1 bin 1, DY
197 MC stat: High-dM DY2M, Nb3plus, recoil bin 2 bin 1, DY
198 MC stat: High-dM QCDCR, Nb1, recoil bin 6 bin 1
199 MC stat: High-dM LLCR, Nb3plus, recoil bin 5 bin 1
200 MC stat: High-dM LLCR, Nb1, recoil bin 3 bin 1
201 MC stat: Low-dM GCR, Nb1, ISR pT >=500, recoil bin 3 bin 1
202 MC stat: High-dM GCR, Nb3plus, recoil bin 1 bin 1
203 MC stat: High-dM LLCR, Nb1, recoil bin 4 bin 1
204 MC stat: High-dM GCR, Nb2, recoil bin 3 bin 1
205 MC stat: High-dM LLCR, Nb1, recoil bin 2 bin 1
206 MC stat: High-dM SR bin 32 bin 1
207 MC stat: High-dM GCR, Nb1, recoil bin 5 bin 1
208 MC stat: Low-dM LLCR, Nb1, ISR pT >=500, recoil bin 2 bin 1
209 MC stat: High-dM SR bin 58 bin 1, signal (1200, 500)
210 MC stat: Low-dM DY2M, Nb1, ISR pT >=500, recoil bin 2 bin 1
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Fit
Pull
+1σ Impact
−1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



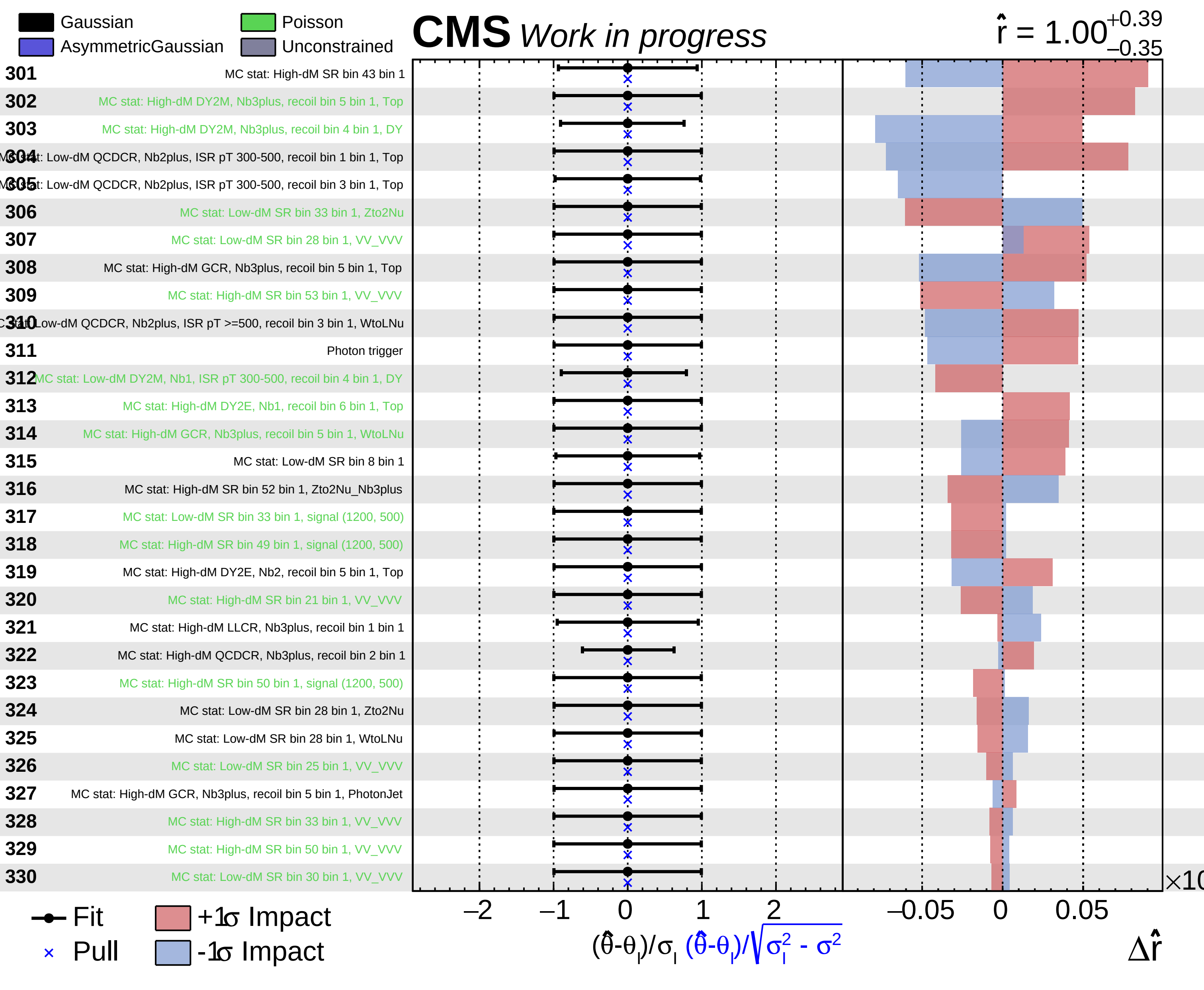
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
211 MC stat: Low-dM GCR, Nb1, ISR pT >=500, recoil bin 4 bin 1
212 Electron trigger
213 MC stat: Low-dM SR bin 15 bin 1
214 MC stat: Low-dM SR bin 25 bin 1, Zto2Nu
215 MC stat: High-dM SR bin 44 bin 1
216 MC stat: Low-dM DY2E, Nb2plus, ISR pT 300-500, recoil bin 2 bin 1
217 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, QCD
218 MC stat: Low-dM LLCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1
219 MC stat: Low-dM SR bin 1 bin 1
220 MC stat: High-dM SR bin 21 bin 1, WtoLNu_Nb1
221 MC stat: Low-dM GCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1
222 MC stat: Low-dM DY2E, Nb1, ISR pT 300-500, recoil bin 3 bin 1
223 MC stat: High-dM SR bin 3 bin 1
224 MC stat: High-dM QCDCR, Nb2, recoil bin 1 bin 1
225 MC stat: High-dM GCR, Nb3plus, recoil bin 3 bin 1
226 MC stat: Low-dM SR bin 32 bin 1
227 MC stat: Low-dM SR bin 2 bin 1
228 MC stat: High-dM DY2M, Nb2, recoil bin 4 bin 1
229 Muon identification
230 MC stat: High-dM SR bin 26 bin 1
231 MC stat: High-dM SR bin 50 bin 1, Top_Nb3plus
232 MC stat: High-dM DY2E, Nb3plus, recoil bin 5 bin 1, DY
233 MC stat: High-dM SR bin 41 bin 1, WtoLNu_Nb2
234 MC stat: Low-dM GCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1
235 MC stat: High-dM GCR, Nb3plus, recoil bin 4 bin 1
236 MC stat: Low-dM LLCR, Nb1, ISR pT 300-500, recoil bin 2 bin 1
237 MC stat: High-dM SR bin 21 bin 1, signal (1200, 500)
238 MC stat: High-dM SR bin 58 bin 1, Zto2Nu_Nb3plus
239 MC stat: High-dM QCDCR, Nb3plus, recoil bin 6 bin 1
240 MC stat: High-dM DY2E, Nb3plus, recoil bin 6 bin 1, DY
−2 −1 0 1 2
−0.5 0 0.5
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



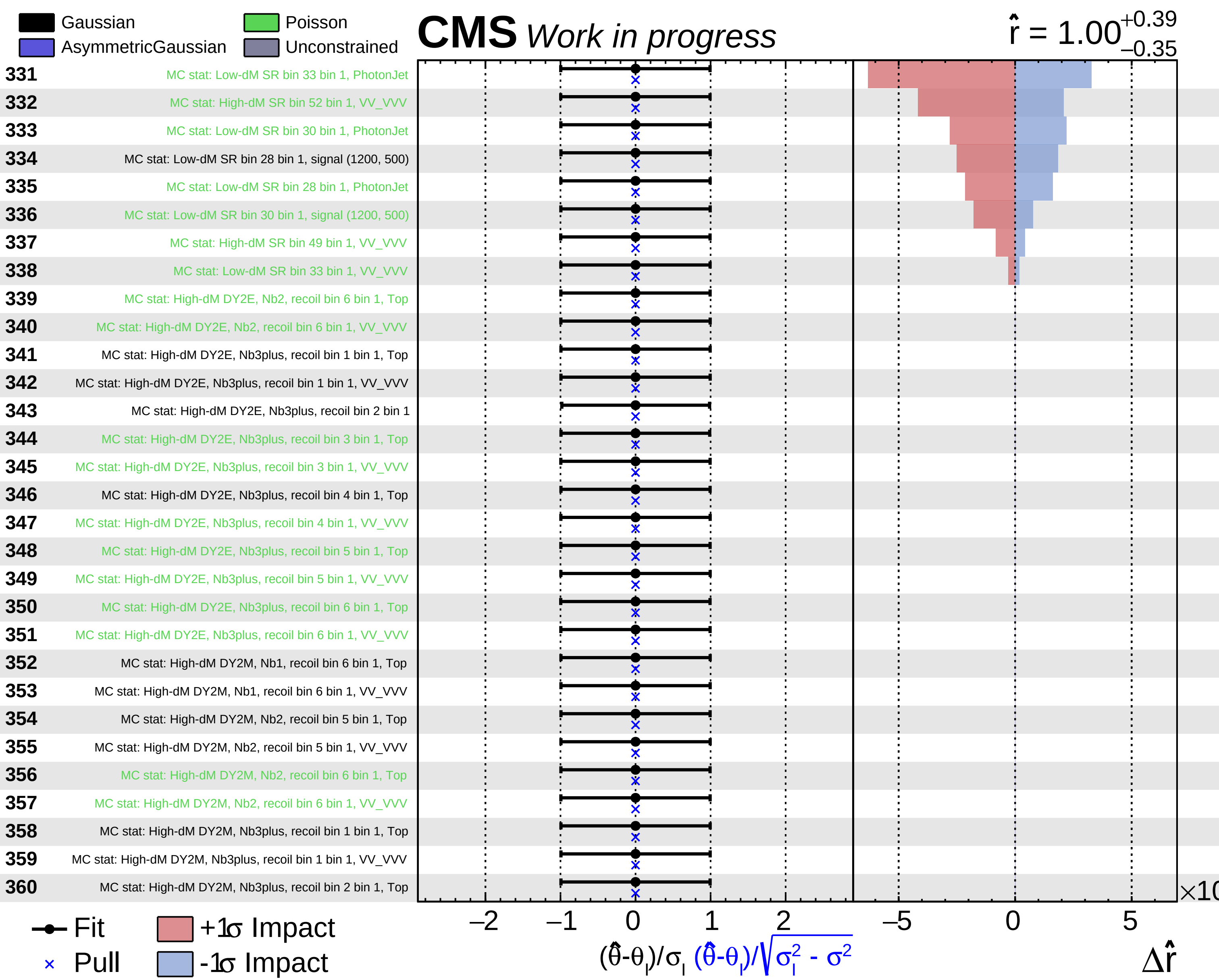
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
241 MC stat: High-dM GCR, Nb2, recoil bin 1 bin 1
242 MC stat: High-dM DY2E, Nb1, recoil bin 5 bin 1
243 MC stat: High-dM DY2M, Nb2, recoil bin 3 bin 1
244 MC stat: High-dM GCR, Nb3plus, recoil bin 2 bin 1
245 MC stat: Low-dM GCR, Nb1, ISR pT 300-500, recoil bin 1 bin 1
246 MC stat: High-dM SR bin 41 bin 1, VV_VVV
247 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 2 bin 1, DY
248 MC stat: Low-dM SR bin 25 bin 1, PhotonJet
249 MC stat: High-dM DY2M, Nb1, recoil bin 5 bin 1
250 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, DY
251 MC stat: Low-dM SR bin 25 bin 1, signal (1200, 500)
252 MC stat: Low-dM SR bin 33 bin 1, Top
253 MC stat: High-dM LLCR, Nb1, recoil bin 6 bin 1
254 MC stat: Low-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 2 bin 1
255 MC stat: Low-dM LLCR, Nb1, ISR pT >=500, recoil bin 4 bin 1
256 MC stat: High-dM GCR, Nb2, recoil bin 6 bin 1
257 MC stat: High-dM QCDCR, Nb3plus, recoil bin 3 bin 1
258 MC stat: Low-dM SR bin 33 bin 1, WtoLNu
259 MC stat: Low-dM GCR, Nb1, ISR pT >=500, recoil bin 2 bin 1
260 MC stat: High-dM SR bin 25 bin 1
261 Muon trigger
262 MC stat: High-dM SR bin 49 bin 1, Top_Nb3plus
263 MC stat: Low-dM QCDCR, Nb1, ISR pT >=500, recoil bin 1 bin 1
264 MC stat: Low-dM SR bin 23 bin 1
265 MC stat: High-dM SR bin 59 bin 1, Zto2Nu_Nb3plus
266 MC stat: Low-dM SR bin 4 bin 1
267 MC stat: High-dM DY2M, Nb3plus, recoil bin 6 bin 1, DY
268 MC stat: High-dM SR bin 33 bin 1, WtoLNu_Nb2
269 MC stat: High-dM SR bin 4 bin 1
270 MC stat: High-dM SR bin 45 bin 1
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



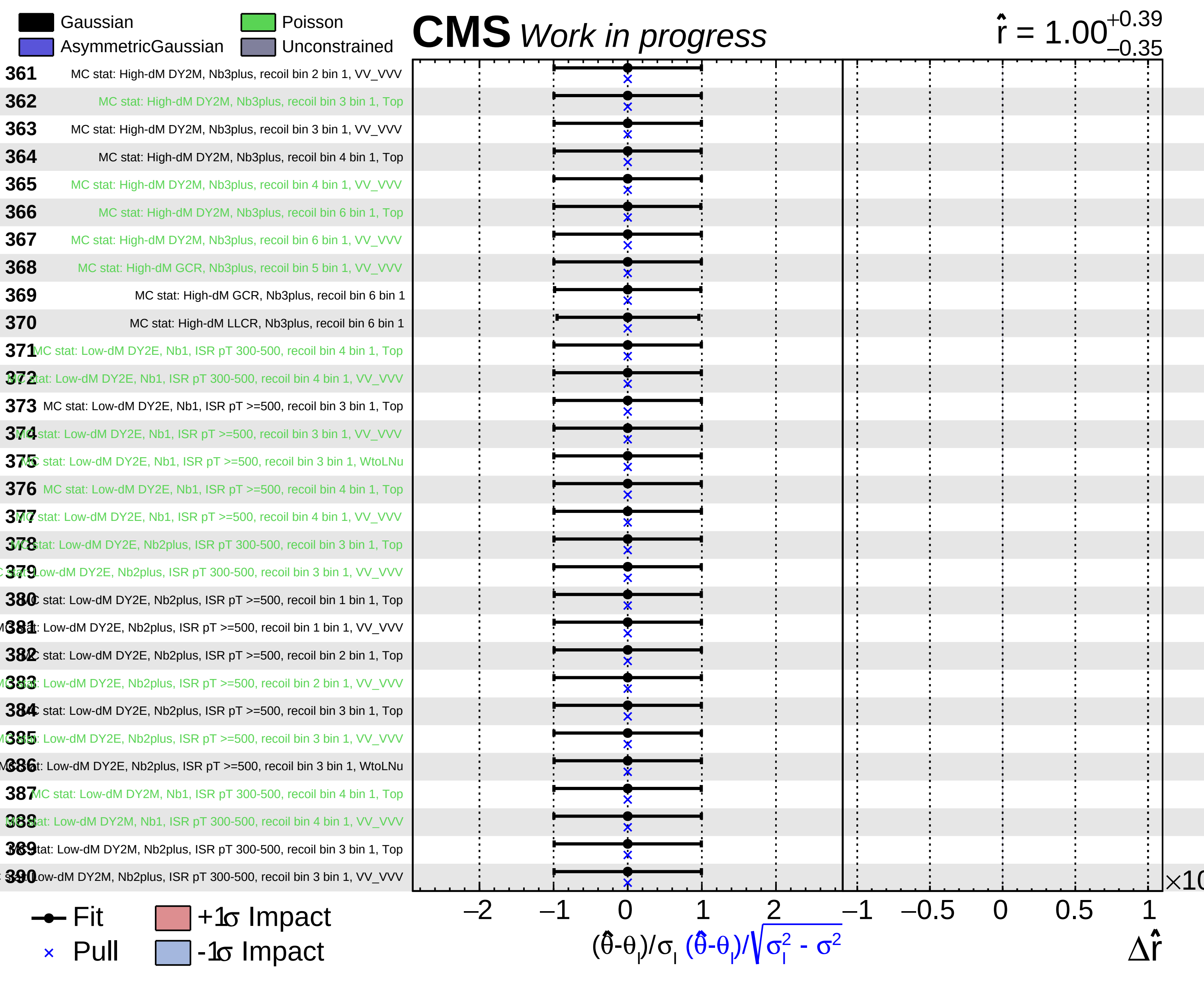
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
271 MC stat: Low-dM LLCR, Nb1, ISR pT >=500, recoil bin 3 bin 1
272 MC stat: Low-dM SR bin 12 bin 1
273 MC stat: Low-dM SR bin 16 bin 1
274 MC stat: High-dM DY2M, Nb2, recoil bin 6 bin 1, DY
275 MC stat: Low-dM SR bin 30 bin 1, Top
276 MC stat: High-dM DY2E, Nb1, recoil bin 6 bin 1, VV_VVV
277 MC stat: High-dM SR bin 33 bin 1, Zto2Nu_Nb2
278 tat: Low-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, DY
279 MC stat: Low-dM QCDCR, Nb1, ISR pT >=500, recoil bin 3 bin 1
280 MC stat: High-dM DY2M, Nb3plus, recoil bin 3 bin 1, DY
281 MC stat: High-dM DY2E, Nb3plus, recoil bin 3 bin 1, DY
282 ow-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, WtoLNu
283 C stat: Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, DY
284 MC stat: High-dM SR bin 33 bin 1, signal (1200, 500)
285 MC stat: High-dM SR bin 52 bin 1, signal (1200, 500)
286 MC stat: High-dM SR bin 57 bin 1, signal (1200, 500)
287 tat: Low-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, Top
288 MC stat: High-dM DY2M, Nb1, recoil bin 6 bin 1, DY
289 MC stat: Low-dM SR bin 28 bin 1, Top
290 MC stat: Low-dM DY2M, Nb1, ISR pT >=500, recoil bin 4 bin 1, Top
291 ow-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, Zto2Nu
292 MC stat: Low-dM LLCR, Nb1, ISR pT 300-500, recoil bin 4 bin 1
293 Low-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, QCD
294 MC stat: Low-dM QCDCR, Nb1, ISR pT 300-500, recoil bin 3 bin 1
295 MC stat: Low-dM SR bin 30 bin 1, WtoLNu
296 MC stat: High-dM LLCR, Nb3plus, recoil bin 4 bin 1
297 MC stat: Low-dM SR bin 30 bin 1, Zto2Nu
298 MC stat: Low-dM GCR, Nb1, ISR pT 300-500, recoil bin 4 bin 1
299 MC stat: High-dM SR bin 51 bin 1, signal (1200, 500)
300 MC stat: High-dM DY2E, Nb2, recoil bin 5 bin 1, VV_VVV
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



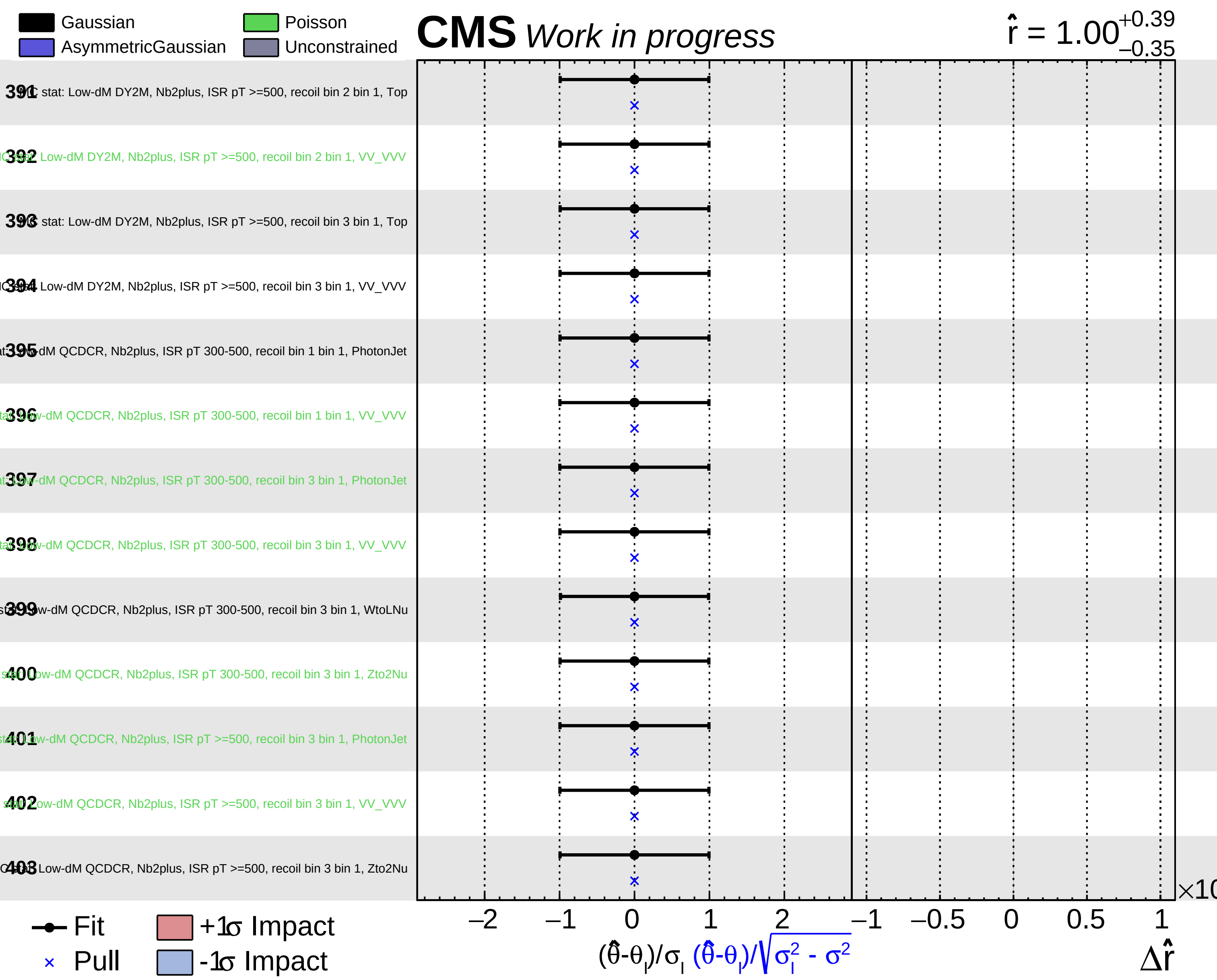
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
301 MC stat: High-dM SR bin 43 bin 1
302 MC stat: High-dM DY2M, Nb3plus, recoil bin 5 bin 1, Top
303 MC stat: High-dM DY2M, Nb3plus, recoil bin 4 bin 1, DY
304 t: Low-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, Top
305 t: Low-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, Top
306 MC stat: Low-dM SR bin 33 bin 1, Zto2Nu
307 MC stat: Low-dM SR bin 28 bin 1, VV_VVV
308 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, Top
309 MC stat: High-dM SR bin 53 bin 1, VV_VVV
310 Low-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, WtoLNu
311 Photon trigger
312 MC stat: Low-dM DY2M, Nb1, ISR pT 300-500, recoil bin 4 bin 1, DY
313 MC stat: High-dM DY2E, Nb1, recoil bin 6 bin 1, Top
314 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, WtoLNu
315 MC stat: Low-dM SR bin 8 bin 1
316 MC stat: High-dM SR bin 52 bin 1, Zto2Nu_Nb3plus
317 MC stat: Low-dM SR bin 33 bin 1, signal (1200, 500)
318 MC stat: High-dM SR bin 49 bin 1, signal (1200, 500)
319 MC stat: High-dM DY2E, Nb2, recoil bin 5 bin 1, Top
320 MC stat: High-dM SR bin 21 bin 1, VV_VVV
321 MC stat: High-dM LLCR, Nb3plus, recoil bin 1 bin 1
322 MC stat: High-dM QCDCR, Nb3plus, recoil bin 2 bin 1
323 MC stat: High-dM SR bin 50 bin 1, signal (1200, 500)
324 MC stat: Low-dM SR bin 28 bin 1, Zto2Nu
325 MC stat: Low-dM SR bin 28 bin 1, WtoLNu
326 MC stat: Low-dM SR bin 25 bin 1, VV_VVV
327 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, PhotonJet
328 MC stat: High-dM SR bin 33 bin 1, VV_VVV
329 MC stat: High-dM SR bin 50 bin 1, VV_VVV
330 MC stat: Low-dM SR bin 30 bin 1, VV_VVV
−2
−1
0
1
2
−0.05
0
0.05
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
331 MC stat: Low-dM SR bin 33 bin 1, PhotonJet
332 MC stat: High-dM SR bin 52 bin 1, VV_VVV
333 MC stat: Low-dM SR bin 30 bin 1, PhotonJet
334 MC stat: Low-dM SR bin 28 bin 1, signal (1200, 500)
335 MC stat: Low-dM SR bin 28 bin 1, PhotonJet
336 MC stat: Low-dM SR bin 30 bin 1, signal (1200, 500)
337 MC stat: High-dM SR bin 49 bin 1, VV_VVV
338 MC stat: Low-dM SR bin 33 bin 1, VV_VVV
339 MC stat: High-dM DY2E, Nb2, recoil bin 6 bin 1, Top
340 MC stat: High-dM DY2E, Nb2, recoil bin 6 bin 1, VV_VVV
341 MC stat: High-dM DY2E, Nb3plus, recoil bin 1 bin 1, Top
342 MC stat: High-dM DY2E, Nb3plus, recoil bin 1 bin 1, VV_VVV
343 MC stat: High-dM DY2E, Nb3plus, recoil bin 2 bin 1
344 MC stat: High-dM DY2E, Nb3plus, recoil bin 3 bin 1, Top
345 MC stat: High-dM DY2E, Nb3plus, recoil bin 3 bin 1, VV_VVV
346 MC stat: High-dM DY2E, Nb3plus, recoil bin 4 bin 1, Top
347 MC stat: High-dM DY2E, Nb3plus, recoil bin 4 bin 1, VV_VVV
348 MC stat: High-dM DY2E, Nb3plus, recoil bin 5 bin 1, Top
349 MC stat: High-dM DY2E, Nb3plus, recoil bin 5 bin 1, VV_VVV
350 MC stat: High-dM DY2E, Nb3plus, recoil bin 6 bin 1, Top
351 MC stat: High-dM DY2E, Nb3plus, recoil bin 6 bin 1, VV_VVV
352 MC stat: High-dM DY2M, Nb1, recoil bin 6 bin 1, Top
353 MC stat: High-dM DY2M, Nb1, recoil bin 6 bin 1, VV_VVV
354 MC stat: High-dM DY2M, Nb2, recoil bin 5 bin 1, Top
355 MC stat: High-dM DY2M, Nb2, recoil bin 5 bin 1, VV_VVV
356 MC stat: High-dM DY2M, Nb2, recoil bin 6 bin 1, Top
357 MC stat: High-dM DY2M, Nb2, recoil bin 6 bin 1, VV_VVV
358 MC stat: High-dM DY2M, Nb3plus, recoil bin 1 bin 1, Top
359 MC stat: High-dM DY2M, Nb3plus, recoil bin 1 bin 1, VV_VVV
360 MC stat: High-dM DY2M, Nb3plus, recoil bin 2 bin 1, Top
−2 −1 0 1 2
−5 0 5
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
$\hat{r} = 1.00^{+0.39}_{-0.35}$
361 MC stat: High-dM DY2M, Nb3plus, recoil bin 2 bin 1, VV_VVV
362 MC stat: High-dM DY2M, Nb3plus, recoil bin 3 bin 1, Top
363 MC stat: High-dM DY2M, Nb3plus, recoil bin 3 bin 1, VV_VVV
364 MC stat: High-dM DY2M, Nb3plus, recoil bin 4 bin 1, Top
365 MC stat: High-dM DY2M, Nb3plus, recoil bin 4 bin 1, VV_VVV
366 MC stat: High-dM DY2M, Nb3plus, recoil bin 6 bin 1, Top
367 MC stat: High-dM DY2M, Nb3plus, recoil bin 6 bin 1, VV_VVV
368 MC stat: High-dM GCR, Nb3plus, recoil bin 5 bin 1, VV_VVV
369 MC stat: High-dM GCR, Nb3plus, recoil bin 6 bin 1
370 MC stat: High-dM LLCR, Nb3plus, recoil bin 6 bin 1
371 MC stat: Low-dM DY2E, Nb1, ISR pT 300-500, recoil bin 4 bin 1, Top
372 MC stat: Low-dM DY2E, Nb1, ISR pT 300-500, recoil bin 4 bin 1, VV_VVV
373 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 3 bin 1, Top
374 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 3 bin 1, VV_VVV
375 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 3 bin 1, WtoLNu
376 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 4 bin 1, Top
377 MC stat: Low-dM DY2E, Nb1, ISR pT >=500, recoil bin 4 bin 1, VV_VVV
378 MC stat: Low-dM DY2E, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, Top
379 MC stat: Low-dM DY2E, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, VV_VVV
380 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 1 bin 1, Top
381 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 1 bin 1, VV_VVV
382 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 2 bin 1, Top
383 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 2 bin 1, VV_VVV
384 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, Top
385 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, VV_VVV
386 MC stat: Low-dM DY2E, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, WtoLNu
387 MC stat: Low-dM DY2M, Nb1, ISR pT 300-500, recoil bin 4 bin 1, Top
388 MC stat: Low-dM DY2M, Nb1, ISR pT 300-500, recoil bin 4 bin 1, VV_VVV
389 MC stat: Low-dM DY2M, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, Top
390 MC stat: Low-dM DY2M, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, VV_VVV
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Work in progress
r̂ = 1.00 +0.39 −0.35
391 stat: Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 2 bin 1, Top
392 Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 2 bin 1, VV_VVV
393 stat: Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, Top
394 Low-dM DY2M, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, VV_VVV
395 -dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, PhotonJet
396 -dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 1 bin 1, VV_VVV
397 -dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, PhotonJet
398 -dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, VV_VVV
399 w-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, WtoLNu
400 ow-dM QCDCR, Nb2plus, ISR pT 300-500, recoil bin 3 bin 1, Zto2Nu
401 ow-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, PhotonJet
402 ow-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, VV_VVV
403 Low-dM QCDCR, Nb2plus, ISR pT >=500, recoil bin 3 bin 1, Zto2Nu
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I (θ̂-θ_I)/√(σ_I² - σ²)
Δr̂